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What do we really know? The effect of reporting thresholds on inferences using environmental right-to-know data

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Abstract
Environmental right-to-know regulations require regulated entities to publicly disclose measures of environmental performance but exempt entities from these disclosure requirements if they manufacture, process, or use a chemical below some threshold level. Environmental right-to-know data are widely used to assess environmental performance by academics, regulators, non-profit organizations, and the public. This paper uses data from Massachusetts to estimate the effect of reporting thresholds in environmental right-to-know programs on the validity of inferences using data from these programs. The analysis indicates that errors in inference introduced by reporting thresholds may be significant. Up to 40% of the observed decline in reported toxic releases in Massachusetts may be attributed to non-reporting due to the reporting thresholds. In addition, quartile rankings of facilities may be in error up to 45% of the time when behavior around the reporting thresholds is not taken into account.

Keywords: environmental data, toxics release inventory, toxics use reduction act, state regulation.

1. Introduction

Information disclosure has been increasingly used as a form of regulation in many policy spheres (Sunstein 1999). Information disclosure regulations take a variety of forms. Some regulations focus on product labeling requirements, for example, cigarette warnings (Fenn *et al.* 2001; Sloan *et al.* 2002); nutritional labeling (Foster & Just 1989; Brown & Schrader 1990); and mercury and fish consumption advisories (Shimshack *et al.* 2007). Other regulations focus on requirements to disclose known information about a transaction to all stakeholders; for example, SEC financial disclosure requirements (Healy & Palepu 2001; Greenstone *et al.* 2006). Still other regulations focus on the public's right-to-know about a broad range of potential health and safety hazards, for example requirements for community water suppliers to disclose information on chemicals in drinking water (Bennear & Olmstead 2008) and requirements for manufacturing facilities to disclose releases of toxic chemicals (Hamilton 2005). Assessing the value of these information disclosure regulations requires understanding what the information being disclosed really means, and what can be properly inferred from it. This paper focuses on

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assessing inferences using one particular type of environmental right-to-know data from the US.

Environmental right-to-know data, in particular data from the US Toxics Release Inventory (TRI), are some of the mostly widely available and widely used data on environmental performance by facilities and firms. This use is due, in part, to the fact that data from right-to-know programs are arguably the most comprehensive data on plant-level environmental performance as they contain data on a variety of chemical releases to all media (e.g. air, water, landfills, underground injection, transfers off-site) (Bennear & Coglianese 2005).

These data have been used by academics as measures of environmental performance in research on topics ranging from voluntary programs (Khanna & Damon 1999; Gamper-Rabindran 2006), environmental management systems (Potoski & Prakash 2005a, b), self-regulation (King & Lenox 2000), state regulation (Bui & Mayer 2003; Helland & Whitford 2003; Bennear 2007), environmental justice (Boer *et al.* 1997; Arora & Cason 1998; Sadd *et al.* 1999; Shapiro 2005), corporate structure (Grant & Jones 2003), as well as research on the impacts of the right-to-know programs themselves (Hamilton 1995; Konar & Cohen 1997; Khanna *et al.* 1998; Fung & O'Rourke 2000; Karkkainen 2001; Hamilton 2005).

Data from right-to-know laws have been used by regulators to target enforcement, inspection, and outreach activities (for example pollution prevention assistance) at either high-priority geographic areas or high priority facilities (US EPA OEI 2003). Non-governmental organizations (NGOs) use the data to target their efforts and to inform the public about environmental hazards in their neighborhoods (US EPA OEI 2003).

Because environmental right-to-know data are so widely used it is important to understand what we can actually "know" about environmental performance based on these data. These right-to-know programs are often complicated regulations with many nuances in terms of what exactly has to be reported and by which facilities. One such nuance under the TRI program is that a facility has to report releases of a chemical only if the facility uses, manufactures, or processes the chemical above some threshold level. The existence of reporting thresholds can complicate inference using right-to-know data because facilities may stop reporting releases for some chemicals in some years simply because in those years the facility was below the use, manufacturing, or processing threshold and therefore not required to report. If facilities are not reporting on the same set of chemicals in every year, inferences based on changes in reported releases over time (at the facility level, the firm level, or by geographic area) can be biased. Similarly, inferences based on relative comparisons of reported release levels among a set of facilities, firms, or geographic areas may be biased.

The impact of the reporting thresholds on inference using TRI data has not previously been assessed because the TRI program does not provide any information on why facilities stop reporting for a chemical. Without such information, it is impossible to ascertain whether release reporting stopped because a facility reduced its use, manufacturing, or processing below the reporting threshold or because the facility stopped using the chemical altogether. This paper is the first to empirically quantify the potential impact of reporting thresholds on the validity of inferences based on changes over time (time series inference) and based on comparisons among entities in one time period (cross-sectional inference). The analysis provided here uses data from one state program, the Massachusetts Toxics Use Reduction Act (TURA), because this state program is the only program to provide sufficient data to identify facilities that stop

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reporting because they are below the reporting threshold. The evidence from Massachusetts indicates that the reporting thresholds do impact reporting behavior for many facilities. Nearly two-thirds of facilities that stop reporting for a listed chemical do so because they use the chemical in quantities below the reporting threshold, but greater than zero. In addition, this change in reporting induced by the thresholds substantially impacts the validity of inferences using the data collected under the program. Estimates of the decrease in toxic chemical releases in Massachusetts between 1990 and 1999 are off by up to 40% in the worst-case scenario. That is, up to 40% of the observed decrease in releases in Massachusetts may be artificial declines created by the reporting thresholds. For cross-sectional comparisons, the results for Massachusetts indicate that quartile rankings of facilities may be wrong as often as 45% of the time when reporting thresholds are not accounted for. Because the Massachusetts program is structured similarly to the federal TRI program the findings for Massachusetts indicate that inferences using the federal TRI data may also be biased by the reporting thresholds.

2. Environmental right-to-know data

2.1. Background

In December 1984, a Union Carbide pesticide plant in Bhopal, India accidentally released 40 metric tons of methyl isocyanate, killing an estimated 2,000 people and injuring 100,000 others. Shortly thereafter, a pesticide release in West Virginia hospitalized 150 people. Partly in response to concerns raised by these high-profile accidental releases, Congress passed the Emergency Planning and Community Right to Know Act (EPCRA) in 1986. EPCRA requires manufacturing facilities to report releases of specific toxic chemicals on an annual basis, and to make these reports available to the public. The US Environmental Protection Agency's (EPA) implementation of EPCRA resulted in the creation of the Toxic Release Inventory (TRI), which requires large manufacturing facilities to publicly disclose total releases of listed toxic chemicals to all media on an annual basis.

There are three factors that determine whether a facility is subject to the TRI disclosure requirements. The first is the industrial sector. Originally, only manufacturing facilities were subject to the TRI reporting requirements. Subsequently, several other industrial sectors were added, including federal facilities, metal and coal mining, electrical generating facilities that combust coal or oil, chemical wholesale distributors, petroleum terminals and bulk storage facilities, and hazardous waste treatment storage and disposal facilities. The final two determinants of regulatory eligibility concern facility size. Only facilities with 10 or more full-time equivalent employees are required to report pollution levels. In addition, facilities are required to disclose releases only of listed chemicals for which they either: (i) manufacture or process more than 25,000 pounds; or (ii) use more than 10,000 pounds.[1] Notice that the reporting thresholds are not based on the actual levels of chemicals released into the environment, but rather on the level of chemical manufactured, processed, or used. A facility that uses 10,001 pounds of Chemical X and releases 80 pounds of it to the environment is required to report those 80 pounds, but a facility using 9,999 pounds of Chemical X and releasing 8,000 pounds of it to the environment is not required to disclose those releases. Figure 1 provides a flow chart of TRI reporting requirements.

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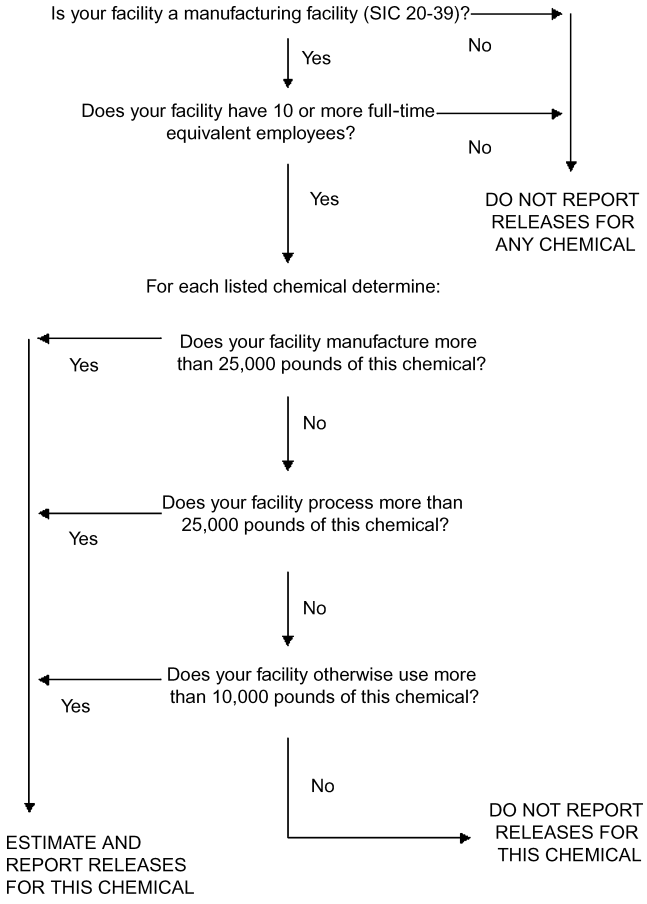

Figure 1 Flow chart for the TRI reporting requirements.

EPCRA requires facilities to disclose toxic releases to the federal EPA and also to their state government. A few states have required additional public disclosure for plants in their jurisdictions. One such state is Massachusetts. In 1989, Massachusetts passed the Toxics Use Reduction Act (TURA), which expanded the reporting requirements for facilities in that state (Massachusetts Department of Environmental Protection, Bureau of Waste Prevention 2003). The Massachusetts program is particularly interesting for this analysis because it collects additional data that allow for an assessment of the impact of the reporting thresholds on inference using right-to-know data.

There are two main differences between the TURA and the TRI requirements that affect analysis of the impact of reporting thresholds on inferences using right-to-know data. The first difference is that the reporting thresholds are weakly lower in Massachusetts. If a facility triggers the federal reporting threshold for at least one listed chemical (i.e. manufacturers or processes more than 25,000 pounds or otherwise uses more than 10,000 pounds), the facility must report for all listed chemicals for which total manufacture, process, and use is greater than 10,000 pounds. Therefore, for most facilities in Massachusetts there are not three separate reporting thresholds, but one binding threshold at 10,000 pounds of total manufacture, process, or use for each listed chemical.

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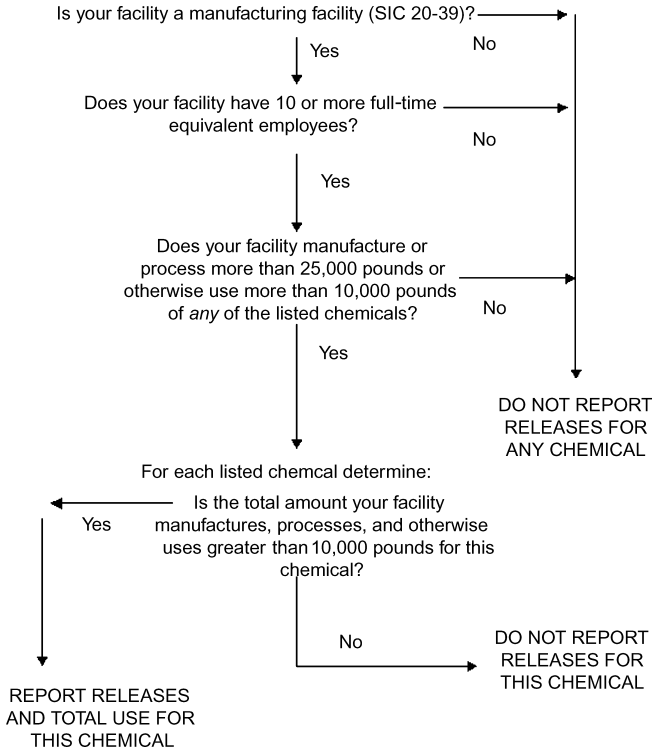

Figure 2 Flow chart for the TURA reporting requirements.

The final difference is that facilities in Massachusetts are required to report chemical manufacture, process, and use amounts in addition to chemical release amounts. As with the federal TRI data, the Massachusetts TURA data have reporting thresholds that are not based on the amount of a chemical released to the environment, but rather on the amount of that chemical that is manufactured, processed, or used at the facility. Unlike the federal program, Massachusetts requires facilities to report total manufacture, process, and use amounts and their total releases of the chemical. However, this disclosure is required only if the facility exceeds the manufacturing, process, and use thresholds. Figure 2 provides a flow chart for the TURA reporting requirements.

If we compare Figure 2 with Figure 1, the top two disclosure requirements based on industry and employees are the same. However, for TURA the reporting requirements based on manufacturing, process, and use amounts differ from the federal regulation. Once a facility manufactures or processes more than 25,000 pounds or uses more than 10,000 pounds of any chemical, the facility must report for all chemicals with total use, process, or manufacture amounts greater than 10,000 pounds.

2.2. Concerns about environmental right-to-know data

Despite the frequent use of right-to-know data in research, there are several known concerns about the validity of these data as measures of environmental performance (US GAO 1991; Natan & Miller 1998; Graham & Miller 2001; US EPA OEI 2002; Toffel & Marshall 2004; de Marchi & Hamilton 2006). The first concern is that the environmental right-to-know data capture only a portion of a facility's environmental performance. Originally data were collected for manufacturing facilities and for approximately 300

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chemicals. The fact that the TRI data are not comprehensive may reduce the significance of these data as a measure of environmental performance. This concern was recognized early in the program (US GAO 1991) and the program has since been expanded to more than double the number of chemicals and to include several additional categories of facilities.

A second known concern with environmental right-to-know data arises from the fact that these data are reported in total pounds of a chemical released, but it is widely acknowledged that pounds of different chemicals present widely different levels of risk (Toffel & Marshall 2004). Some of the factors that determine the overall environmental risk presented by a facility's activities include the total pounds of releases, the toxicity of these releases, the exposure level in the population, the duration of exposure, and the sensitivity of the population. Simply adding up total pounds of different chemicals released and comparing across facilities may provide an incorrect ordering of facilities according to relative risk. EPA and academic researchers have been working to develop indicators that take the TRI data as an input, combine it with toxicity, exposure, and degradability information, and then calculate a risk-based metric. EPA has released its Risk Screening Environmental Indicators (RSEI) model, which combines TRI data and risk information to develop a risk-adjusted pollution measure (US EPA OPPT 2007). Recent academic work has analyzed the validity of several different weighting mechanisms (Toffel & Marshall 2004). Using these weighting schemes will doubtless improve the quality of estimates of environmental risk using right-to-know data.

A third concern stems from the fact that right-to-know data are self-reported by facilities. Changes in how a facility estimates releases or even how it classifies releases may lead to changes in reported releases that are not reflective of real changes in environmental performance (Natan & Miller 1998; Graham & Miller 2001). Recent research compares self-reported data from the TRI with monitoring data for air pollutants and finds that for lead and nitric acid the self-reported data are significantly different from the monitoring data (de Marchi & Hamilton 2006).

A final concern arises from the existence of reporting thresholds. As discussed in the previous section, not all facilities are required to disclose releases of every listed chemical. Facilities are required to disclose releases only if they employ 10 or more full-time equivalent employees and if they exceed certain reporting thresholds. Moreover, these reporting thresholds are not based on chemical releases, but rather on chemical manufacture, process, and use amounts. This last concern has generally been overlooked in the literature and it is the magnitude of the bias in inference caused by these thresholds that is the focus of this paper.

The reporting thresholds create bias in inference using TRI data if facilities sometimes, but not always, report releases of a particular chemical. This could occur for any number of reasons, such as natural variation in production levels, a facility changing manufacturing processes for unrelated reasons, or a reduction in the amount of a chemical used in the wake of the adoption of a new technology. It could also occur because facilities strategically manage their operations to ensure that they can be classified as below the reporting threshold for as many chemicals as possible.

Regardless of the reason for periodic reporting, the lack of consistent reporting for a chemical can bias inferences based on environmental right-to-know data. For example, imagine a facility initially uses 10,500 pounds and releases 1,000 pounds of Chemical X and also uses 20,000 pounds and releases 5,000 pounds of Chemical Y. Because the

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facility uses more than 10,000 pounds of both chemicals it must report releases for both chemicals, and thus total reported releases at the plant are 6,000 pounds. Then, in the next year, the facility uses 9,999 pounds and releases 950 pounds of Chemical X and uses 20,000 pounds and releases 5,000 pounds of Chemical Y. Because the facility does not use more than 10,000 pounds of Chemical X, the facility is not required to disclose those 950 pounds of releases. Total reported releases at the facility in this year are only 5,000 pounds – when actual releases were 5,950 pounds. This difference in reporting affects researchers' and policymakers' inferences about what happened to the facility over time (time series inference) and also can affect inferences of how well the facility is doing relative to other facilities (cross-sectional inference).

In this paper I provide upper and lower bounds on the bias in both the time series and cross-sectional inferences that could be attributed to changes in reporting induced by the reporting thresholds. I first assess how frequently facilities stop reporting for a chemical because they go below the reporting thresholds and then assess how this behavior affects the use of these data for research.

3. Methods

To assess whether or not there is significant variation in reporting from year to year based on the reporting thresholds, there must be some data on facilities that stop reporting releases of listed chemicals. The Massachusetts Toxics Use Reduction Act (TURA) is the only state or federal right-to-know program that contains enough information on facilities that stop disclosing releases to make reasonable estimates of the magnitude of the bias in both time series and cross-sectional inference induced by the reporting thresholds. I will refer to this type of bias in inference as "reporting threshold bias." The TURA data provide two sources of variation that allow for the identification of bounds on reporting threshold bias.

First, if a facility does not report for a chemical in one year for which it had reported in the previous year, the TURA reporting forms include an optional question that asks the facility to explain the change. Approximately one-third of facilities that cease reporting for a chemical answer the optional question. The second source of variation in the TURA data that can be used to help estimate bounds on the reporting threshold bias is that facilities are required to report how much of a chemical they use in addition to how much they release. The federal TRI has reporting thresholds based on use, but requires reporting only on releases. We might reasonably expect that a facility is more likely to go below the reporting threshold if it is close to it, but with the federal data we cannot calculate how close a facility is to the manufacture, process, or use thresholds. The presence of the chemical manufacture, process, and use data in the TURA program allows for calculation of the distance from the reporting threshold for each facility and each chemical.

The additional data from the Massachusetts TURA program allow for estimation of bounds on the degree of reporting threshold bias. There are three steps to the estimation. In the first step, I impute the reason a facility stopped reporting for all facilities that do not directly provide a reason for this behavior by answering the optional question.

In the second step, I use three different procedures to estimate the magnitude of "missing" releases for facilities that stopped reporting because they went below the reporting threshold. The first procedure leads to a lower bound estimate on total releases

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at the facility. The lower bound estimate assumes that when a facility ceases reporting its true releases are zero. This is the implicit assumption currently made by government agencies, non-profits, and academic researchers when aggregating release data to the facility level (or higher levels). For example, in its analysis of the most recent year of TRI data, EPA claims that "[f]rom 2005 to 2006, total disposal or other releases on- and off-site decreased by 105 million pounds or 2%" (US EPA 2008), a claim necessarily based on the assumption that releases of all non-reported chemicals are zero. The second procedure results in what can be considered an upper bound estimate. In this upper bound scenario, if a facility ceases to report a chemical because it went below the reporting threshold, releases are set equal to the most recent level of releases reported for that chemical at that facility. Moreover, the facility is assumed to continue releasing the chemical at the same level in perpetuity. The third procedure leads to an estimate of missing releases that extrapolates the value of non-reported releases based upon linear trends in reported releases. These three procedures, then, generate an upper and a lower bound and an intermediate estimate of the magnitude of releases not reported due to the reporting thresholds.

The third step to the overall estimation strategy in this paper involves assessing the effect of these missing releases on trends in releases over time and on cross-sectional comparisons of facilities within a given year. For policy purposes it is not enough to show that missing releases are significant. Rather one is interested in whether these missing releases substantially alter interpretations of either the changes in environmental performance over time or relative rankings of facilities within a given time period.

Figure 3 diagrams the estimation strategy. In Figure 3, "other reason" means any reason for not reporting other than going below the reporting threshold. A description of those other reasons appears in Section 4.1.

4. Findings

4.1. Assessing and imputing reasons facilities stop reporting

Facilities in Massachusetts were required to disclose pollution and chemical use data to the state TURA program beginning in 1990. From 1990 to 1999, there were a total of 23,200 chemical reports filed by 1,092 facilities. During this same time period, there were 3,758 cases where a facility reported for a chemical in one year, but did not report the chemical in the following year. For these facilities, the TURA form provides an optional question where the facility can explain why it is no longer reporting for this chemical. The question is multiple-choice with the following six possible responses: (i) chemical use is below the reporting threshold but greater than zero; (ii) the chemical was not used that year; (iii) a different chemical was substituted; (iv) chemical use was eliminated without substitution; (v) decline in business; or (vi) other (Massachusetts Department of Environmental Protection, Bureau of Waste Prevention 2003). If the facility answers "other," it is given the option to fill in a reason. This question is answered for a total of 1,271 (or 33.8%) of the cases where reporting ceases. Table 1 provides the distribution of responses.

Approximately two-thirds of all respondents to the optional question answered that they were no longer reporting because their chemical use was below the reporting threshold, but greater than zero. Of the respondents that answered "other," the most

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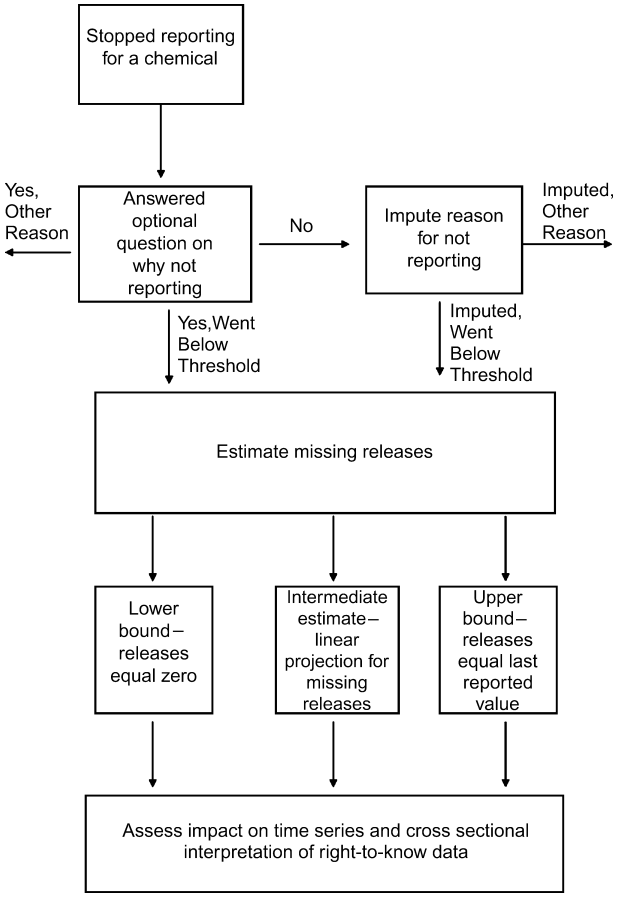

Figure 3 The estimation strategy.

frequent explanation was that the chemical in question had been delisted by the state and reporting was no longer legally required.

The high percentage of respondents that state that they ceased reporting because their chemical use was positive, but lower than the reporting threshold, raises concern that the degree of reporting threshold bias may not be trivial. However, the optional question was only answered by one-third of the relevant observations. It is necessary to impute a reason for not reporting for the two-thirds of observations that did not answer the optional question. Before conducting the imputation, it is worth determining whether the facilities that responded to the optional question are systematically different from those that did not respond. If the facilities that respond to the optional question are not significantly different from those that do not respond, then the answers to the optional question can be thought of as coming from a randomly selected subset of facilities. Inferences from that randomly selected subset are likely to be representative of the population as a whole. If the group that answers the optional question is not a random subset of the whole population of TURA reporters, then some adjustments will be needed in making inferences between the sample that answered and the sample that did not answer.

To see whether or not the responders to the optional question are a representative sample, I compare the distribution of the data for the two groups for three

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Table 1 Explanation for non-reporting among respondents to the optional question in Massachusetts

Reason for no longer reporting	Number of respondents	Percentage of all non-missing responses
Chemical use is below the reporting threshold but greater than zero	844	66.40
Chemical was not used this year	96	7.55
Substituted a different chemical	87	6.85
Chemical use eliminated without substitution	60	4.72
Decline in business	31	2.44
Other	153	12.04

variables – two-digit SIC code, year reporting stopped, and total releases to the environment in the year before reporting stopped.[2] The data suggest that responders to the optional question are not systematically different from non-responders on total pounds of chemicals released. The difference in the average release levels is 1,353 pounds with a standard error of 4,567 pounds. There are some differences in industry and the year that reporting stopped. Of the 28 industrial sector (SIC) codes, the two groups – responders and non-responders – are statistically different for seven of them.[3] There are also some differences among responders and non-responders in the years for which reported ceased. Of the nine years, the distribution of responders and non-responders differs in four.[4] In addition, one may be concerned that there are unobservable differences between the responders and the non-responders that also are correlated with whether or not the facility goes below the reporting threshold, but still uses the chemical in positive quantities. In the absence of a valid instrument that can explain that a facility chooses to answer the optional question, but does not directly explain that facility going below the threshold, specific sample selection correction models cannot be employed. Rather, in this section I impute reasons for non-reporting for those that did not answer the optional question based on data from the sample of facilities that answered the optional question, controlling for total releases, industry, and year. The differences in industry and year between those that answered the optional question and those that did not are thereby accounted for in the imputation method. I also conduct sensitivity analysis to assess how much the results of the main analysis vary with changes in the imputation method. The sensitivity analysis can be found in Section 4.4.

The first step in the imputation method is to create a new dummy variable that takes a value of one if the facility ceased reporting because it went below the reporting threshold and zero otherwise. I then use this variable as the dependent variable in a logit estimation on observable facility characteristics. One characteristic that may explain the propensity to go below the reporting threshold is the facility's distance from that threshold. To measure the distance from the reporting threshold, I construct a variable that measures, for each chemical, the difference between actual chemical manufacture, process, or use amounts (in pounds) and the reporting threshold.[5] The hypothesis is that the greater the distance from the reporting threshold, the less likely a facility is to cease reporting that chemical because its use of the chemical went below the reporting threshold. Similarly, if there is a relationship between chemical use and chemical release, then total releases (in pounds) of the chemical in time t may predict

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Table 2 Results for estimation of threshold logit for respondents to optional question

	Coefficients
Constant	1.02 (1.19)
Distance from threshold	−0.00001 (0.000003)*
Releases	−0.00003 (0.000006)*
SIC dummies	Yes†
Year dummies	Yes†

*Significant at the 1% level.
†An F-test shows variables are jointly significant at the 1% level. Number of observations: 1,251. Pseudo R^2 0.17.

whether the facility goes below the reporting threshold in year $t + 1$. Thus, lagged total releases are also used as an explanatory variable. Other potential explanatory variables include industry dummy variables, and year dummy variables that proxy for changes in industry best practices and exogenous technological change. The relationship estimated is then given by:

$$\text{below_threshold}\begin{pmatrix}0\\1\end{pmatrix}_{ict} = \alpha + \beta_1 \text{Distance from Threshold}_{ict-1} + \beta_2 \text{releases}_{ict-1} + \beta_{sic} SIC_{it} + \beta_t Year_t + \varepsilon_{ict}$$

where subscript i indicates facility, c indicates chemical, and t indicates year.

The results of this estimation are present in Table 2.[6] As expected, a facility is more likely to go below the reporting threshold for a chemical if it is closer to the reporting threshold or had lower releases of the chemical.

The above equation was used to predict the probability of going below the threshold for the 2,482 observations with no answer to the optional question. Observations with a predicted probability greater than 50% were coded as going below the threshold. The choice of a probability cutoff is a balance between a Type 1 and Type 2 error. Choosing a lower cutoff ensures that more observations that actually went below the reporting threshold are classified that way, but there are more "false positives" – observations that did not go below the reporting threshold but are misclassified as having done so. Choosing a higher cutoff has the opposite problem – there will be fewer false positives but more false negatives (facilities that really did go below the reporting threshold but are not classified that way).

In this case, we can test the rate of false positives and false negatives by comparing assignment based on different probability cutoffs with the actual answer to the optional question for the set of facilities that answered the question. With the 50% cutoff, 790 of 829 facility-chemical-year observations (95%) that actually went below the reporting threshold are classified correctly, but more than 50% of observations that did not go below the reporting threshold are misclassified as having done so. In other words, with a 50% cutoff there are almost no false negatives but a number of false positives. If the cutoff is increased to 75%, only 423 of the 829 facility-chemical-year observations (51%) are correctly classified (higher rate of false negatives), but the rate of false positives drops to 16%. Using the 50% cutoff means little true reporting threshold bias will be missed, but the estimates are likely to be overinclusive and can be thought of as worst-case scenarios on the magnitude of the reporting threshold bias. In Section 4.4, I present

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Table 3 Distribution of observations (facility-chemical-year) that cease reporting by explanation

	Explanation provided by facility	Explanation predicted	Total
Below the reporting threshold	844	1,786	2,630
All other reasons	424	696	1,120
Total	1,268	2,482	3,750

sensitivity analysis in order to assess how much the results of the analysis vary when the percent cutoff is varied between 50% and 100%.

Table 3 provides a breakdown of the observations that ceased reporting by explanation. Of the 2,482 observations for which no explanation for non-reporting was provided, 1,786 (72%) were predicted to have stopped reporting because the facility went below the reporting threshold for that chemical, but still use the chemical in positive quantities. This is very similar to the percentage of observations that actually reported that they went below the reporting threshold. Of the 1,268 observations with an answer to the optional question, 844 (66%) went below the reporting threshold.

In summary, analysis of the Massachusetts TURA data indicates that a substantial percentage of facilities that cease reporting a chemical do so because they go below the reporting threshold for that chemical, but still use it and may still have positive releases of these chemicals. While the frequency with which facilities stop reporting because they go below the reporting threshold seems to present some concern about the validity of inferences using environmental right-to-know data, the level of concern may still be low if the total amount of releases that disappear from the registry is small. That is, perhaps the percentage of observations affected by the reporting thresholds is large, but the total share of releases represented by these observations is small. The next section addresses the question of the likely magnitude of missing releases.

4.2. Estimating the magnitude of missing releases

The difficulty in assessing the effect of the reporting thresholds on the validity of the right-to-know data is daunting. It is impossible to know how large a problem non-reportable releases are, precisely because these releases are no longer reported. The best we can do is to assess how large this problem might reasonably be, given observable information. To that end, this section focuses on estimating the magnitude of "missing" releases by estimating bounds on the possible size of these releases – a lower bound estimate, an upper bound estimate, and an intermediate (linear projection) estimate.

To make these results meaningful, I restrict the analysis to chemicals and industries that have been subject to the TURA reporting requirements since its inception – the so-called core group. This ensures that we are examining variations in reported releases due to the reporting thresholds and not changes in releases due to changes in the regulatory requirements themselves. There were also no changes to the reporting thresholds for these chemicals during the time frame used in this analysis.

I perform my analysis for two different measures of pollution from the TRI. The first is on-site releases. On-site releases are releases of the pollutant to air, water, land, or underground injection that occur at the facility's location. The second measure is total on- and off-site releases. Total releases include on-site releases, but add transfers of waste to off-site locations for disposal or recycling. In general, it is thought that total releases

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are overinclusive as a measure of environmental performance because some of the transfers are not pollution, but are transfers for recycling and reuse. On-site releases, however, are underinclusive as a measure of environmental performance because some waste that is transferred off-site is pollution attributable to the facility. In at least one case, research has indicated that facilities have responded to an environmental program by decreasing on-site releases and increasing off-site transfers (Gamper-Rabindran 2006). It is important to see if the reporting thresholds have different effects for the two measures.

In addition, this analysis focuses on the raw sum of releases across chemicals. In Section 2.2, I discussed concerns about using raw pounds as a measure of environmental performance and different weighting schemes that have been developed to account for toxicity and exposure. However, the majority of research to date, both by academics and governments, has relied on unweighted sums of pounds released (Hamilton 1995; Boer *et al.* 1997; Konar & Cohen 1997; Arora & Cason 1998; Khanna *et al.* 1998; Khanna & Damon 1999; Sadd *et al.* 1999; Fung & O'Rourke 2000; Karkkainen 2001; Bui & Mayer 2003; Helland & Whitford 2003; Bennear 2007). These analyses may also be biased by not accounting for toxicity, but those biases are better understood (Toffel & Marshall 2004). The purpose of this paper is to isolate and quantify the potential bias in existing analyses caused by the reporting thresholds alone and doing that requires focusing on the measures actually used in those analyses, which are unweighted sums of releases.

The lower bound estimate of the value of non-reported releases is that these releases are zero. This assumes that once a facility drops below the reporting threshold for a given chemical, it no longer releases any of that chemical. A conservative upper bound estimate of the value of non-reported releases is that they equal the last reported value of releases for that chemical at that facility. For example, if a facility reports uses 10,001 pounds and reports releases of 500 pounds of a chemical in year t, and then drops below the reporting threshold by using 9,999 pounds of that chemical in year $t + 1$, the upper bound estimate of missing releases is that this facility releases 500 pounds a year of that chemical in perpetuity.[7]

Providing upper and lower bounds on the potential reporting threshold bias provides useful information on how large the bias might be. However, it does not provide any information on the probability distribution of the true bias within that range. While this range is useful, perhaps we would wish to also have an estimate of bias under a more probable scenario. An alternative assumption about the behavior of releases for non-reported chemicals is that these releases will follow a similar pattern to releases for reported chemicals. If there are exogenous changes (such as technological change) that result in decreases in releases over time, then these changes are likely to apply to non-reported releases as well. Under this assumption, one can project trends in releases for non-reportable data based on observed trends in releases among reported chemicals.

The linear best predictor of reported releases turns out to be a distributed lag of past releases. A simple one-period distributed lag, where current releases of each chemical are regressed on releases of that chemical from the previous year, explains 80% of the variation in releases. Increasing the number of lags increases the predictive power slightly, but makes the equation less useful for prediction because a smaller number of observations have multiple lags. Using the one-period distributed lag model, current on- and off-site releases are, on average, 95% of previous years' releases. Similarly, using

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Table 4 Estimates of TURA Releases in Massachusetts (in 1,000s of pounds)

Year	Lower bound = reported releases	Linear projection	Upper bound	% difference between lower and upper bounds	% difference between lower bound and linear projection
Total reported releases (on- and off-site)					
1991	43,372	45,291	45,392	4.7	4.4
1992	44,870	48,495	48,777	8.7	8.1
1993	38,199	43,696	44,245	15.8	14.4
1994	38,833	45,141	45,998	18.5	16.2
1995	36,557	43,493	44,692	22.3	19.0
1996	36,984	44,308	45,864	24.0	19.8
1997	32,449	39,848	41,775	28.7	22.8
1998	33,048	40,898	43,228	30.8	23.8
1999	19,967	28,282	31,032	55.4	41.6
On-site releases					
1991	17,653	18,915	19,230	8.9	7.1
1992	15,573	17,517	18,288	17.4	12.5
1993	11,862	14,598	16,015	35.0	23.1
1994	9,300	12,100	14,170	52.4	30.1
1995	8,715	11,238	13,886	59.3	28.9
1996	7,451	9,855	13,061	75.3	32.3
1997	6,367	8,553	12,271	92.7	34.3
1998	5,621	7,697	11,905	111.8	36.9
1999	5,126	7,113	11,786	129.9	38.8

a one-period distribution lag model, current on-site releases are 80% of previous years' releases.

Using the coefficients from the one-period distributed lag model, I predict releases for each facility that ceases reporting. This generates trends that are downward sloping for all facilities that cease to report for a chemical; the rate of decrease is larger for on-site releases than for total releases. Table 4 provides estimates of the magnitude of missing releases, both total releases (on- and off-site) and on-site releases, using both the upper bound assumption and the linear projection.

The results indicate that the degree of missing releases in the early years is relatively modest as a percentage of total releases. In 1991, missing releases are 4–5% of total (on- and off-site) releases and between 7 and 9% of reported on-site releases. However, over time, missing releases as a percentage of total reported releases rise dramatically. There are two reasons for this. First, missing releases are cumulative. In every year, about 2–4% of the previous year's releases are not reported due to facilities going below the reporting threshold. But in each year, the total amount of missing releases are all those releases that are no longer being reported by all facilities; these include the releases from facilities that stopped reporting for the chemical this year, as well as the missing releases from facilities that went below the reporting threshold two years ago, three years ago, and so forth. Particularly in the upper bound scenario, when facilities that drop below the reporting threshold are assumed to continue releasing at the same level in perpetuity, the cumulative effect can be quite large.

The second explanation is that reported releases fall considerably over time, even for facilities that continue to report for all chemicals. For the upper bound case, where facilities

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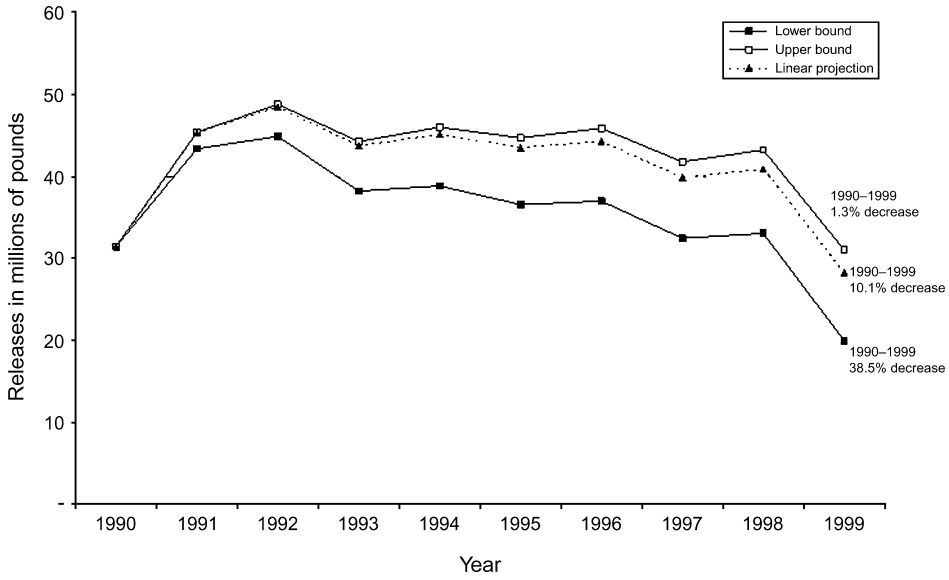

Figure 4 The trend in total releases over time for each of three scenarios.

are assumed to continue to release non-reported chemicals at the same level forever, the relative importance of these releases increases over time – the missing releases stay the same, but the total reported releases decrease – resulting in a sharp increase in missing releases as a percentage of total reported releases. In fact, for on-site releases the difference between the upper and lower bound estimates of releases differs by 130% in 1999.

4.3. Assessing the policy impact of missing releases

The data in Table 4 illustrate that the magnitude of missing releases generated by the existence of the reporting threshold is non-trivial. But do these missing releases significantly bias estimates in the trend in environmental performance over time? Figure 4 graphs the trend in total releases over time for each of three scenarios: (i) only reported releases (lower bound estimate, assumes non-reported releases equal zero); (ii) reported releases plus the upper bound estimate of missing releases (sets missing releases equal to their last reported value for all years); and (iii) reported releases plus linearly extrapolated estimates of missing releases.

It is clear that the estimate of the trend in environmental performance is substantially affected by the exclusion or inclusion of the estimated releases for non-reported chemicals. If one uses a lower bound assumption, that all non-reported releases are zero, the change in total releases from 1990 to 1999 is 36.7%. However, if one uses instead the upper bound assumption, that facilities that fall below the reporting threshold continue to release the same amount forever, the change in total releases over the same 10-year period was only 0.6%. If one uses the more moderate linear projection, the change in total releases over the 10-year period is 10.1%.

The story is somewhat less stark, but still significant, if one ignores the upward trend in reported off-site releases between 1990 and 1991 and instead examines trends only for the years 1991 to 1999. For this period, the lower bound assumption leads to the conclusion that total releases have fallen by 54.0%, the upper bound assumption estimates the decline at 31.6%, and the linear projection leads to an estimate of a

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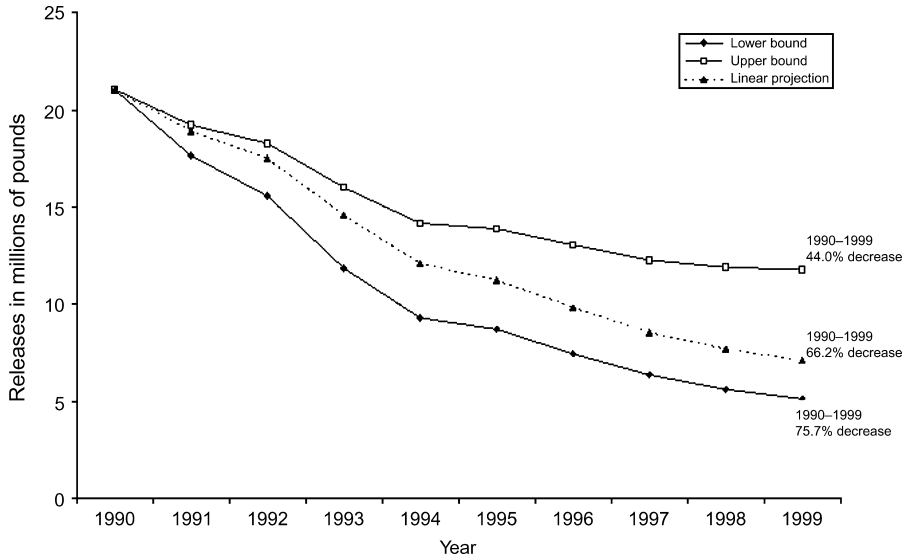

Figure 5 The estimated trends in on-site releases.

37.6% decrease. If one uses the difference in trends from 1991 to 1999, it appears that in the worst-case, non-reported releases by facilities that fall below the reporting threshold may account for approximately 22 percentage points of the total 54 percentage point decrease in reported releases, or roughly 40%.

The potential degree of bias introduced by the reporting thresholds in the trends for on-site releases is similar. Figure 5 provides the estimated trends in on-site releases. If one uses the lower bound assumption, the decrease in on-site releases from 1990 to 1999 was 75.7%. If one uses the upper bound assumption, the decrease over this period was 44.0%. Finally, if one uses the assumption of linearly projected decreases in non-reported releases, the change over the decade was 66.2%. Thus, of the observed 75% decline in reported releases, from 1990 to 1999 as much as 31.6 percentage points (or 40%) of this decrease may be accounted for by non-reported releases.

As evidenced by these data, our understanding of the magnitude of the change in environmental performance can be substantially affected by reporting threshold bias. Making different assumptions about what happens to releases when facilities cease to report because they fall below the reporting threshold makes a big difference in our understanding of the level of environmental improvement. That said, it is somewhat reassuring that even in the upper-bound scenario, on-site releases of toxic chemicals have fallen by 44% in Massachusetts. In this case, using TURA data for time series inference is very likely to result in erroneous cardinal estimates of the magnitude of improvement, but unlikely to result in erroneous ordinal estimates – things have improved by somewhere between 44 and 76%.

A separate concern is whether the reporting thresholds induce bias in cross-sectional comparisons of facilities. Imagine that we rank facilities from lowest releases to highest releases. Regulators and NGOs often do this type of ranking for targeting purposes (US EPA OEI 2003). If the degree of non-reported releases is small relative to any given facility's total reported releases, adjusting for these non-reported releases will change our estimate of how much a facility releases, but may not change the facility's rank much.

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Table 5 Quartile ranking comparisons for total releases

		Upper bound ranking			
		1	2	3	4
Lower bound ranking	1	**1,245**	885	799	275
	2	297	**367**	25	4
	3	260	481	**659**	102
	4	0	64	317	**1,413**

If, in contrast, the magnitude of missing releases is large relative to total reported releases for some set of facilities, then adjusting facility totals to account for this could substantially change the rankings of facilities.

To investigate the degree of reporting threshold bias in cross-sectional inferences, I first divide facilities into quartiles based on their reported releases. Thus, facilities whose reported releases are in the lowest 25% of all facilities in that year are assigned to quartile number one. Facilities whose releases are among the highest 25% of all facilities are assigned to quartile number four. These are the quartile rankings associated with the lower bound assumption that all non-reported releases are equal to zero. I then recalculate the facility's quartile ranking based on the upper bound assumption and the linear projection. If the relative importance of the non-reported releases is small, we would expect to see few facility quartile rankings change. In this case, we would not be terribly concerned about the potential bias in cross-sectional inference introduced by the reporting thresholds. However, if the relative importance of missing releases is large, we might expect to see more facility quartile rankings change dramatically. An alternative way to think about this is to imagine assigning a facility a grade based on its pollution levels relative to other facilities in a given year. A facility can get a grade of excellent, good, fair, or poor. If we adjust for missing releases resulting from facilities going below the reporting threshold, do we change our grades for only a few facilities or for a substantial number of facilities? The results of the quartile distribution comparison are provided in Tables 5 and 6 for total and on-site releases, respectively.

In each table, the diagonal elements (in bold) represent facilities whose quartile ranking is unaffected by inclusion of estimates of their missing releases. All off-diagonal entries are facilities whose quartile rankings differ under the upper and lower bound assessments. For total releases, 31% of facilities have different quartile rankings under the upper bound assessment than under the lower bound assessment. For on-site releases 52% of facilities have different quartile rankings under the two different scenarios.

For the purposes of using rankings from environmental right-to-know data for regulatory targeting purposes, perhaps one is most concerned about errors in the top and bottom quartiles. That is, one is concerned most with mislabeling a facility as "poor" or "excellent." Focusing on these quartiles, we can see from the data in Table 5 that 21% of the facilities that are labeled "poor" based on their reported releases would have been labeled either "fair" or "good" once reported releases are adjusted to include an upper bound assessment of missing releases. In addition, 61% of facilities that were labeled "excellent" using only reported releases would have been labeled "good," "fair," or "poor" if missing releases are included. The results for on-site releases are quite similar (Table 6). For on-site releases, 33% of facilities labeled "poor" using reported releases should have received a higher grade, and 58% of facilities labeled "excellent" should have received a lower grade.

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Table 6 Quartile ranking comparisons for on-site releases

		Upper bound ranking			
		1	2	3	4
Lower bound ranking	1	**1,430**	629	864	515
	2	233	**321**	30	6
	3	170	640	**491**	70
	4	0	176	415	**1,230**

This potential variation in ordinal rankings of facilities within a given year is large enough to be of substantial concern – we are potentially assigning the wrong grade two-thirds of the time in the best quartile and a quarter of the time within the worst quartile. If regulatory or enforcement resources are targeted based on rankings of reported releases, these resources are likely to be misallocated.

4.4. Sensitivity analysis

One of the key components of the above analysis was imputing whether or not a facility ceased reporting because it went below the reporting threshold if the facility did not answer the optional question. As discussed above, the choice of a probability cutoff for the imputation was a balance between false positives and false negatives. How sensitive are the findings to changes in the imputation method?

The first sensitivity analysis is to increase the probability cutoff from the logit model from 50 to 75%. Increasing the cutoff to 75% increases the rate of false negatives, meaning that some observations that really went below the reporting threshold will be missed. This will lower the estimate of the reporting threshold bias. The second sensitivity analysis is to examine only missing releases for facilities that actually respond to the optional question and state that they cease reporting because they went below the reporting threshold, but still use the chemical in positive quantities. No imputation is made for facilities that do not answer the optional question. This is essentially the same as increasing the probability cutoff from the logit to 100%. This is the most conservative estimate of the reporting threshold bias.

The third analysis does not designate a cutoff, but rather weights releases by the predicted probability that the facility went below the threshold from the logit. Thus if the logit yields a predicted probability that a facility went below the reporting threshold, but still uses the chemical of 20%, then the upper bound estimate of that facility's releases is 20% of the last reported value in perpetuity.

Table 7 presents the results of the sensitivity analysis for estimates of the effect on the trend in total and on-site releases. For comparison, the first column contains the original estimates (where the probability cutoff equals 50%). Increasing the cutoff from 50 to 75% decreased the magnitude of the reporting threshold bias by 40 to 45%. This difference is not surprising because the numbers from using the 50% cutoff had few false negatives but many false positives, and therefore were likely to overstate the magnitude of the bias. The estimates using the 75% cutoff had significant false negatives, but few false positives and therefore were likely to understate the magnitude of the bias. The "true" amount of bias is unknowable, but likely to fall somewhere in between these two estimates. Under the most conservative scenario, which used data only from facilities

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Table 7 Maximum percent of observed decline in releases from 1991 to1999 that could be explained by reporting threshold bias

	50%	75%	100% (responders only)	Weighted by probability from logit
Total releases	41.4	22.3	13.1	37.3
On-site releases	45.4	25.9	14.6	40.1

Table 8 Maximum potential percent error in on-site cross-sectional rankings due to reporting threshold bias

	50%	75%	100% (responders only)	Weighted by probability from logit
Total	52	42	28	51
Bottom quartile	33	21	14	30
Top quartile	58	48	30	59

that actually answered the optional question, up to 13% of the observed decrease in total releases and up to 15% of the observed decrease in on-site releases from 1991 to 1999 could be due to missing releases. Using the probabilities from logit to weight missing releases yields an estimate between the 50% and 75% estimate, but much closer to the 50% estimate. The important take-away message is that under all scenarios there is a significant percentage of the decrease in reported releases that is attributable to reporting threshold bias.

Sensitivity analysis was also done on the cross-sectional rankings. Table 8 presents the results for the differences between the upper bound and lower bound quartile rankings for all three modifications to the probability cutoff (75%, 100%, and expected value). The first column of the table provides the main findings based on a 50% cutoff for comparison. In the most conservative scenario, the quartile rankings are wrong about 28% of the time. The rankings are wrong only about 14% in the bottom quartile. That is, only about 14% of facilities that would have received a label of "poor" using the reported data should have received a higher grade if missing data were incorporated. In the upper quartile, the percent error is still higher than for the whole sample. Approximately 30% of facilities that would have been labeled "excellent" based on reported releases would have received a lower ranking if missing releases had been accounted for.

Overall, the sensitivity analysis supports the general findings presented in the main analysis. First, a potentially significant share of the decrease in observed releases may be due to facilities no longer being legally obligated to report releases because their use of the chemical is below the reporting threshold. Second, the reporting thresholds may also skew the cross-sectional rankings of facilities. In particular, facilities that appear to be low releasers or good environmental performers based on TURA releases may actually not be better than other facilities with higher releases. The error in cross-sectional ranking diminishes as one moves down the distribution. The rankings are considerably less wrong about identifying the worst facilities.

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5. Policy implications for using environmental right-to-know data

The above analysis indicates that changes in reporting in response to regulatory thresholds may introduce substantial bias in both the time series and cross-sectional estimates of environmental performance using TURA data. Because federal and other state right-to-know programs also have regulatory reporting thresholds, these findings have implications for the use of all environmental right-to-know data, including the frequently used Toxics Release Inventory data. The analysis was restricted to Massachusetts because this is the only state that has data that allow for estimation of the bounds on the potential bias created by changes in reporting around the reporting thresholds.

There are important implications of this finding for at least three groups of environmental right-to-know data users. First, for researchers in both academia and government who often rely on these data as measures of environmental performance outcomes, this research indicates that the number of chemicals reported by a facility should be treated as an additional policy outcome.[8] When investigating the effectiveness of a new regulation, it is not sufficient to examine only the effects on total reported releases (even if weighted by toxicity) because if the policy has an effect on the number of facilities that reduce chemical use below the reporting threshold, then estimates of the policy's effect on releases are likely to be biased upward. Ignoring changes in reporting around the reporting threshold will overstate the effect of the policy on environmental performance. If, however, one estimates that the policy does not have an effect on the number of chemicals reported, but does have an effect on total releases, one can feel more confident that the policy has actually improved environmental performance and not simply reduced reporting.

The second group of data users that are affected by the findings in this study are administrators of right-to-know programs. Changing reporting thresholds in right-to-know programs is currently an active area of regulatory activity. In 2000, EPA lowered the reporting thresholds for mercury from the 25,000/10,000 pound standard threshold to 10 pounds, and in 2001 EPA lowered the reporting threshold for lead from the 25,000/10,000 pound standard threshold to 100 pounds. For mercury, the lower reporting thresholds resulted in a 20-fold increase in the number of facilities reporting mercury releases and a 33% increase in total reported releases of that chemical. For lead, the decrease in the reporting threshold resulted in a fourfold increase in the number of facilities reporting and a 19% increase in the total amount of lead releases reported. There has been substantial debate on whether or not the benefits of this additional reporting outweigh the costs associated with assisting these new reporters and managing the additional data. This paper highlights additional benefits to expanding reporting requirements. Increasing the comparability of data over time and across facilities in a given year can allow for better targeting of limited enforcement and compliance resources and allow for better assessment of progress in improving environmental performance. Efforts to change reporting requirements should aim to decrease reporting threshold bias, or at least take care not to increase such bias and thereby further limit the usefulness of this important data resource.

The third group affected by these findings is other government and NGO users who rely on environmental right-to-know data for targeting purposes. This includes government agencies that rely on these data to target inspection, enforcement, and outreach activities either to specific facilities or to specific geographic regions. NGOs also use

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right-to-know data for targeting purposes. The results of this analysis suggest that identifying the "best" and "worst" facilities can be sensitive to reporting a threshold bias. At a minimum, if one observes large changes in reported releases one should carefully consider the number of chemicals reported. If large decreases in releases correspond to decreases in chemicals reported, these changes should be viewed more cautiously. A more sophisticated model of targeting could use methods similar to those employed here to estimate non-reported releases and see how the targeting efforts would differ if these releases are included. The good news for this particular use of the data is that errors comparing cross-sections of facilities tend to be smaller when the goal is identifying the worst facilities rather than identifying the best facilities. This suggests that targeting uses of environmental right-to-know data would be better focused on identifying poor performers and not used as a means of identifying environmental leaders.

Since the inception of the TRI program in 1987, these data have been widely used as measures of environmental performance. They have also been widely criticized as biased measures of performance. This paper provides the first quantified evidence of a type of bias that has heretofore been unmeasurable – the bias induced by arbitrary reporting thresholds. This bias can be significant both for assessing improvements in environmental performance over time and for assessing relative environmental performance among entities in a single time period. However, this analysis should not be interpreted as a condemnation of inferences using environmental right-to-know data, which remain one of the few sources of information on environmental releases to all media by US facilities, but rather as a caution that such inferences must carefully account for this particular nuance of the regulation that generates these data.

Notes

1 EPA has lowered these reporting thresholds for lead, mercury, and other bioaccumulating toxins in recent years, but for the time period under study the reporting thresholds for all chemicals were unchanged and were based on 25,000 pounds of manufacturing or processing or 10,000 pounds of use.

2 These variables are the only variables available in the TURA data that are reasonably comparable among those that answer the optional question and those that do not. Other variables that might be of interest such as size of the facility and age of the facility are not available.

3 A Pearson's χ^2-test rejects equality of the industry distributions across the two groups.

4 A Pearson's χ^2-test rejects equality of the year distributions across the two groups.

5 The distance from the reporting threshold should always be positive because reporting is only required for facilities that are above the threshold. However, for 213 of 23,200 facility-chemical-year observations the facility reported for a chemical even if it was below the reporting threshold. This amounts to less than 1% of all observations and inclusion or exclusion of these anomalous observations does not affect the results.

6 Other specifications were run, in particular specifications including the square of the distance to the threshold. Inclusion or exclusion of the distance squared variable made no difference in the predicted probabilities from the logit regressions and therefore does not affect the remainder of the analysis.

7 Occasionally a facility will report for a chemical for a few years, then stop reporting for that chemical because it is below the reporting threshold, and then begin reporting for the chemical again in later years. In this case, the upper bound estimate of releases equals observed releases in any year in which the facility actually reports releases and in non-reporting years are set equal to the releases in the most recent year in the past for which the facility reported releases for that chemical.

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8 See Bennear (2007) for an example of a policy analysis that uses number of chemicals reported as an additional outcome measure.

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